

MUHAMMAD SYAHMI BIN ABDUL RAHMAN

Engineering Project Manager | Robotics, Automation & Industrial IoT

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PROFESSIONAL SUMMARY

Engineering project manager with 7+ years of experience delivering robotics, autonomous systems, and Industrial IoT (IIoT) integration initiatives from R&D through production deployment. Currently lead a cross-functional team of 6 to 7 engineers, with a proven track record architecting ROS2 and MQTT communication frameworks and delivering automation platforms on time and within budget. Improved engineering delivery efficiency by 20% and increased client retention by 25% through structured delivery processes and technical leadership. Targeting Engineering Manager roles focused on scalable system architecture, team development, and product-driven automation strategy.

CORE COMPETENCIES

Engineering Team Leadership, Robotics & Autonomous Systems, Industrial IoT (IIoT) Integration, System Architecture & Design, ROS2 & MQTT Frameworks, Agile & Sprint Planning, Product Development Lifecycle, Cross-Functional Team Management, Stakeholder Communication, Risk Management & Mitigation, Budget & Resource Planning, PLC Programming, Edge & Embedded Systems, Technical Documentation & SOPs.

PROFESSIONAL EXPERIENCE

Project Manager, R&D and System Integration

Sailcott (M) International Sdn. Bhd. | Shah Alam, Malaysia | 2020 - Present

Lead a cross-functional team of 6 to 7 engineers driving end-to-end development and deployment of robotics and automation systems across R&D and commercial environments.

- Direct development of autonomous mobility platforms including Electric Cart Carrier (ECC) and Autonomous Buggy systems from concept through commissioning.
- Lead cross-functional teams of 6 to 7 engineers spanning software, electrical, and mechanical disciplines to align system architecture and integration milestones.
- Architected ROS2 to MQTT communication framework enabling reliable cloud-to-robot data transmission and scalable IoT integration.
- Delivered 3 major automation projects ahead of schedule while maintaining budget compliance.
- Improved engineering delivery efficiency by 20% through structured sprint planning, workflow optimization, and standardized reporting.
- Reduced integration issues through structured validation, testing, and commissioning protocols.
- Serve as primary technical liaison between engineering teams, management, and clients to align product requirements with deployment timelines.

Automation & IoT Consultant (Freelance, alongside primary role)

Independent Practice | Shah Alam, Malaysia | 2018 - Present

Provide industrial automation, PLC programming, and IoT system integration services to SMEs across manufacturing and energy sectors.

- Delivered 15+ automation and IoT projects including PLC systems, CRM integration, and cloud-based real-time monitoring dashboards.
- Designed MQTT-based IoT architectures and Node-RED dashboards for operational analytics and performance tracking.
- Increased client retention by 25% through proactive system optimization and structured reporting.
- Reduced manual reporting workload by implementing automated monitoring and control systems.

- Managed project scope definition, milestone planning, vendor coordination, and deployment execution end to end.

Automation Project Intern

East Automation & Engineering Sdn. Bhd. | Subang Jaya, Malaysia | 2018

- Supported system design, prototype deployment, and validation for solar monitoring and laser surveillance automation systems.
- Developed GUI control interfaces for automated monitoring solutions.

TECHNICAL SKILLS

Robotics & Integration: ROS2, MQTT, Node-RED, REST API Integration, Industrial IoT Platforms

Programming & Control: Python, C, C++, PLC Programming (Ladder Logic)

Hardware & Edge Systems: NVIDIA Jetson Orin, Raspberry Pi, Embedded Linux, Microcontrollers

Project & Collaboration Tools: Microsoft Project, Jira, Trello, Asana, Git

Documentation & Visualization: Microsoft Office, Visio, Draw.io

Methodologies: Agile, Scrum, Sprint Planning, Risk Management, System Validation & Testing

EDUCATION

Bachelor of Engineering, Electronics Engineering

Universiti Teknologi MARA (UiTM), Shah Alam | Jan 2017 - Jan 2019

Diploma in Electrical Engineering (Power)

Universiti Teknologi MARA (UiTM), Dungun | Jan 2014 - Sep 2016

ADDITIONAL INFORMATION

Languages: Bahasa Malaysia (Native), English (Professional Proficiency)

References: Available upon request.